

# SARTHAK MAITY

✉ sarthakmaity450@gmail.com    🌐 sarthak-802.github.io    📞 +91 79081 11197

📍 Dhasachandpur, Mansuka, Ghatal, Paschim Medinipur, West Bengal, 721212, India

## EDUCATION & TRAINING

**Indian Institute of Technology Indore (IIT Indore)**

Aug 2023 – May 2025

M. Sc. in Chemistry (CGPA: 8.57/10.00)

Advisor: Prof. Biswarup Pathak

Theoretical & Computational Chemistry

**Midnapore College (Autonomous)**

Sep 2020 – Jun 2023

B. Sc. in Chemistry (HONS) (CGPA: 8.00/10.00)

**Baradongal Ramanath Institution (10+2)**

2018 – 2020

Physics, Chemistry, Mathematics, Biology (PCMB) (Score: 92.4/100)

## SCHOLARSHIPS & AWARDS

Swami Vivekananda Merit-cum-Means Scholarship (SVMCM)

2018 – 2023

## SCHOLASTIC ACHIEVEMENTS

Secured All-India Rank **504** (Chemistry) in Joint Admission Test for Masters (IIT JAM)

2023

## PUBLICATIONS

† co-first authorship; \* corresponding author.

1. **Maity, S.<sup>†</sup>**; Sharma, R.<sup>†</sup>; Minhas, H.; Pathak, B.\* Deciphering Key Descriptors for Scaling Relationships in Graphene-Supported Pt<sub>n</sub> Clusters via Machine Learning. *Small*, **2026**, DOI: 10.1002/sml.202513283

## WORK & RESEARCH EXPERIENCE

### Theoretical & Computational Research

**Department of Chemistry, IIT Indore**

Aug 2024 – May 2025

Computational Material Design Group (CMDG)

Postgraduate Researcher

**Master's Dissertation:** *Understanding ORR Activity of Nanoclusters Electrocatalysts: Combined DFT and Machine Learning Approach for Multiscale Modelling* | [Thesis]

**Research Advisor:** Prof. Biswarup Pathak

- An ensemble-based DFT framework that captures fluxionality by considering LEME of subnano Pt clusters instead of a single static structure.
- An interpretable supervised ML model enabling accurate prediction of ORR intermediate adsorption energies.
- Descriptor-level identification of scaling-relation breakdown using uncertainty quantification, revealing the geometric features that control non-scalable ORR behavior in subnanometer catalysts.
- A coverage-dependent ORR activity analysis up to monolayer coverage, integrating ab initio oxidation thermodynamics and reaction network analysis under electrochemical conditions.

### Project Student

June 2025 – Feb 2026

**M.Sc. Research Continuation & Advanced Training in GNNs, MLIPs & Catalysis**

**Supervisor:** Prof. Biswarup Pathak, IIT Indore, India.

- Developed expertise in ML for materials science, including active learning, t-SNE, and  $\Delta$ -learning.
- Developed practical expertise in MACE and MEGNet for atomistic simulations and materials-property prediction.
- MLIP-based prediction of adsorption energies in high-entropy alloys (HEAs) and nanoclusters.

## Project-II

Oct 2023 – Feb 2024

**Title:** *Superior Anchoring Effect of Cu-Benzenehexathial MOF as an Al-S Battery Cathode Host*

**Supervisor:** Prof. Biswarup Pathak, IIT Indore, India.

- Identified 2D Cu-BHT MOF as an efficient sulfur host cathode via DFT calculations.
- Demonstrated stronger binding of Al polysulfides compared to graphene-based hosts.
- Showed enhanced sulfur incorporation and reduced polysulfide dissolution, improving cycling stability.

## Experimental Research

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### Project-I

Aug 2023 – Oct 2023

**Title:** *Synthesis and Characterization of N-Containing Heterocyclic Compounds*

**Supervisor:** Prof. Umesh A. Kshirsagar, IIT Indore, India.

- Developed synthetic routes for fused bicyclic and heterocyclic compounds relevant to drug design.
- Achieved synthesis of 4H-pyrido[1,2-a]pyrimidin-4-ones using  $\beta$ -keto esters and PPA.
- Performed Cu(I)-catalyzed oxidative synthesis of imidazo[1,2-a]pyridines in aqueous media.

### Project-III

Feb 2024 – May 2024

**Title:** *Sequestration of Radioiodides Using Ionic Porous Organic Polymer*

**Supervisor:** Prof. Suman Mukhopadhyay, IIT Indore, India.

- Designed and synthesized a multifunctional ionic porous organic polymer (iPOP-Bpy).
- Incorporated viologen as a redox-active unit to enhance iodine capture efficiency.
- Achieved high iodine uptake with rapid adsorption kinetics across gas and solution phases.

## Machine Learning Projects

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**Car Price Prediction (2024) | Movie Recommender System (2025) | IPL Win Probability Predictor (2025)**

*Python, Scikit-learn, NumPy, Pandas, Streamlit | GitHub (Deployment pending)*

## RESEARCH INTERESTS

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- **AI  $\times$  Materials:** First-principles-informed machine learning for materials discovery using Generative AI, Graph Neural Networks (GNNs), and universal machine-learned interatomic potentials (uMLIPs), with applications in inverse design, generative modeling, and structure-property mapping.
- **Catalysis:** Exploring catalyst dynamics, reaction pathways, active-site evolution, and structure-activity relationships through data-driven modeling of fluxional clusters, molecules, and electrochemical interfaces.
- **Energy Storage Systems:** Rational design of next-generation Metal-S (Li-S, Al-S), Metal-ion (Li-ion, Na-ion), and Dual-ion (Li-DIB) batteries through the DFT/MD modeling of intercalating ion, electrode design, electrolyte optimization, and electrode-electrolyte interfacial phenomena.

## SKILLS

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- **Atomistic ML Tools:** Machine Learning Interatomic Potentials (MLIPs): MACE, CHGNet, MatGL, ALIGNN
- **Programming Skills:** Neural Networks, Deep Learning (DGL, PyTorch), Machine Learning (Scikit-learn), Python (NumPy, Pandas, Matplotlib, Seaborn), LaTeX, FORTRAN
- **Software:** VASP, VESTA, Chemcraft, Gaussian, GaussView, Materials Studio, OriginPro
- **Office Tools:** MS Word, PowerPoint, Excel
- **Languages:** English, Hindi, Bengali

## CERTIFICATIONS

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- **Supervised Machine Learning**, Coursera, DeepLearning.AI, Credential ID: 6FVINQX6DJK7
- **Advanced Learning Algorithms**, Coursera, DeepLearning.AI, Credential ID: DDCPX6RMMJ9C
- **Neural Networks and Deep Learning**, Coursera, DeepLearning.AI, Credential ID: IPJK7DTZ4WJW

## CONFERENCES & SEMINARS

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|---------------------------------------------------------------------------------|----------|
| • In-house Symposium, Department of Chemistry, IIT Indore                       | Mar 2024 |
| • International Indo–German Conference on Sustainable Chemistry–III, IIT Indore | Mar 2025 |